



UNIVERSITY OFTM
KWAZULU-NATAL

INYUVESI
YAKWAZULU-NATALI

***In vitro* simulation of BV-associated bacterial response to
antibiotics: A phenotypic and genotypic study**

Master of Medical Science (Medical Microbiology)

Project Proposal

Phiwokuhle Benneden Ntuntuza

226217758

Supervisor: Dr Andile Mtshali

Co-Supervisor: Dr Sinaye Ngcapu

CAPRISA Microbiome Laboratory; Department of Medical Microbiology; School of
Medicine; College of Health Sciences; University of KwaZulu-Natal

TABLE OF CONTENT

1. INTRODUCTION.....	3
2. LITERATURE REVIEW.....	4
2.1.1 Overview of bacterial vaginosis and vaginal microbiota dysbiosis	4
2.1.2 Interspecies synergy, biofilm formation, and their contribution to BV persistence.....	6
2.1.3 Metronidazole as Standard Therapy for BV: Mechanism of Action.....	7
2.1.5 Molecular mechanisms of MTZ resistance in anaerobic polymicrobial communities.....	10
2.1.5.1 Nitroreductases, nim genes, and other resistance determinants.....	11
2.1.5.2 DNA repair and efflux pump-mediated resistance	12
2.1.6 In Vivo Studies of BV Treatment Response: Insights and Limitations.....	13
2.2 Problem Statement.....	14
2.3 Aims and Objectives	14
2.3.1 Aim	14
2.3.2 Objectives.....	14
2.3.3 Hypothesis.....	15
3. METHODOLOGY	15
3.2.1 Study design, setting and sample collection.....	15
3.3 Laboratory Procedures.....	16
3.3.2 Simulation of treatment and phenotypic assessment.....	17
3.3.2.1 Molecular and community shifts analysis.....	18
3.3.2.2 DNA extraction.....	18
3.3.2.3 16S rRNA Amplicon Library preparation.....	19
3.3.2.4 Quantitative PCR.....	20
3.3.2.5 Genotypic analysis	21
3.4 Statistical analysis	21
3.4.1 Differential abundance analysis of microbes.....	21
3.4.2 Community shifts and relative abundance	21
4. TIMELINE	23
5. REFERENCES.....	24

1. INTRODUCTION

The vaginal microbiota is a complex polymicrobial environment, which plays a critical role in maintaining the vaginal reproductive health. Under normal physiological conditions, the *Lactobacillus* spp protect vaginal epithelial cells against pathogen colonisation by maintaining an acidic environment, primarily through the production of lactic acid (1). Bacterial vaginosis (BV) is characterised by a shift from a *Lactobacillus*-dominant community to a dysbiotic state, marked by a reduction in *Lactobacillus* spp. and an increased relative abundance of diverse anaerobic bacteria such as *Gardnerella vaginalis*, *Fannyhessea vaginae* and *Prevotella bivia* (2). Among women of reproductive age, BV remains the most common gynaecological condition globally, with an estimated of approximately 30%, although this varies by population and diagnostic criteria. In sub-Saharan Africa, BV prevalence is notably higher, with estimates of 42.1% among women aged 15–24 and 41.2% among those aged 25 and 49 years (3). The exact etiology of BV remains unclear, however factors such as unprotected sexual intercourse, douching, use of antibiotics and intrauterine devices have been reported to induce vaginal dysbiosis. BV is associated with a range of adverse reproductive and gynaecological outcomes such as spontaneous miscarriages, pre-term birth, infertility and increased risk of acquiring sexually transmitted infections (STIs), including HIV, which is already a high burden in Sub-Saharan (4-6). In addition to its clinical consequences, BV can have psychosocial impacts, as symptoms such as vaginal malodour are often associated with feelings of shame, embarrassment and reduced self-esteem among affected women (4, 6)

According to the Centre for Disease Control and Prevention guidelines, metronidazole (MTZ) is recommended as the first line of treatment for symptomatic women with BV.

Recommended regimens include oral administration of 500 mg of MTZ twice daily for 7 days or applying a MTZ gel 0.75% using one full applicator (5g) intravaginally, once daily for 5 days. Despite these treatments, recurrence is common, with up to 50–60% of women experiencing BV recurrence within 6-12 months following therapy (5). The reduced and often transient effectiveness of MTZ has been attributed to multiple factors, including antimicrobial resistance, incomplete eradication BV-associated bacteria (BVAB) and biofilm-induced tolerance (7). Although *in vivo* studies provide important insights into treatment outcomes including resistance patterns (8), they may not fully capture the mechanistic processes underlying treatment failure, and *in vitro* models provide complementary opportunities for controlled mechanistic investigations. Therefore, this study aims to to characterise microbial

community shifts and identify molecular biomarkers associated with MTZ and clindamycin treatment success and failure in an *in vitro* model of BV.

2. LITERATURE REVIEW

2.1.1 Overview of bacterial vaginosis and vaginal microbiota dysbiosis

The vaginal microbiome is a complex microbial ecosystem, consisting of diverse microorganisms such as bacteria, viruses, and fungi, which collectively play a role in maintaining women's reproductive health and disease. In healthy reproductive-aged women, the vaginal microbiota is typically dominated by *Lactobacillus* species which protect the vaginal epithelial cells against pathogenic agents through multiple including, maintaining an acidic vaginal environment (pH <4.5) through production of lactic acid, competitive exclusion of pathogenic microorganisms, and production of antimicrobial metabolites such as bacteriocins (9). Oestrogen further contributes to vaginal homeostasis by stimulating glycogen accumulation within the epithelial cells, which is subsequently released into the vaginal lumen during epithelial cell shedding (1). This glycogen is then broken down into simpler sugars by host derived alpha amylases and serves as a major nutrient source shaping the vaginal community composition and stability (10). An imbalance in this *Lactobacillus*-dominated environment leads to vaginal dysbiosis. Specifically, this occurs when *Lactobacillus spp* decreases due to overgrowth of anaerobic bacteria such as *G. vaginalis*, *F. vaginae* and *P. bivia*. This dysbiotic state is associated with the production of volatile amines, responsible for fishy odour and an increase in the vaginal pH. Clinically, BV remains the most common cause of white-greyish vaginal discharge in women (2, 11, 12). Importantly, BV is increasingly recognized as a polymicrobial condition involving complex bacterial interactions and biofilm formation, particularly involving *G. vaginalis*, which may contribute to treatment failure and recurrence.

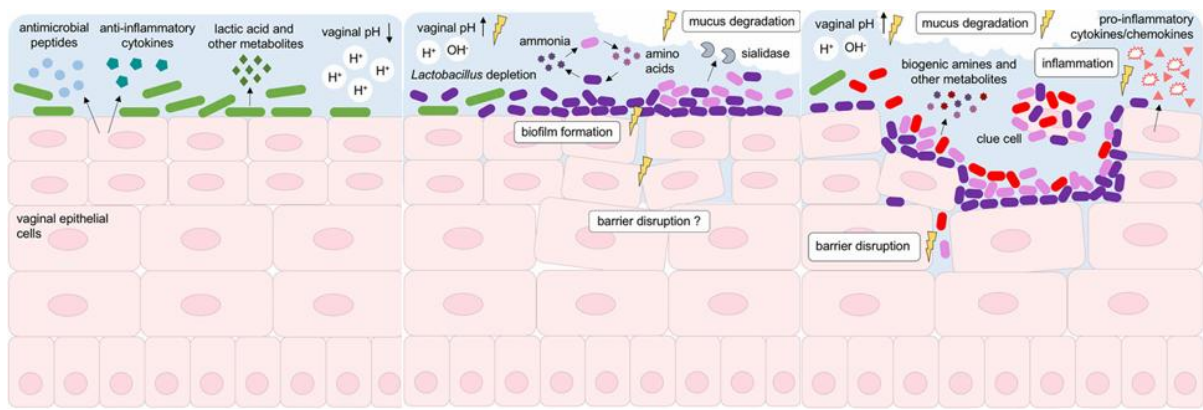


Figure 1: Process of BV pathogenesis from lactic-acid producing *Lactobacillus* to highly diverse strict and facultative anaerobe dominance (13).

Clinically, BV is commonly diagnosed using Amsel's criteria or Nugent scoring (NS). Nugent scoring, which is based on Gram-stained vaginal smears, remains the laboratory gold standard for BV diagnosis. NS values are interpreted as follows: 0-3 indicate BV negative, 4-6 = intermediate BV and NS = 7-10 BV positive (14). Although Nugent scoring is widely used, it has limitations, including inter-observer variability and inability to fully capture the complexity of vaginal microbial communities. Advances in molecular sequencing technologies particularly 16S rRNA gene sequencing, have advanced the understanding of different vaginal microbiome composition and its relationship with BV. These approaches led to the classification of vaginal bacterial communities into five community state types (CSTs) based on the relative abundance of the species dominating (15). CST I, II, III and V are dominated by *L. crispatus*, *L. gasseri*, *L. iners* and *L. jensenii*, respectively. CST IV is further sub-classified into CST IV-A, CST IV-B, CST IV-C, and CST IV-D (16). In contrast, CST IV is characterized by low abundance of *Lactobacillus* spp. and increased diversity of anaerobic BV-associated bacteria (BVAB), including *G. vaginalis*, *Atopobium* species, and *Prevotella* species, and has been further subdivided into CST IV-A and CST IV-B (17, 18). Unlike the gut microbiota, increased microbial diversity within the vaginal microbiota is generally associated with dysbiosis and adverse reproductive health outcomes. Furthermore, the role of *L. iners* remains controversial, as it is frequently detected in both healthy and dysbiotic vaginal states and is considered less stable than *L. crispatus*-dominated communities. Consequently, CST III is often regarded as a transitional or unstable state that may predispose women to progression toward CST IV-associated dysbiosis.

Although the precise aetiology of BV remains incompletely understood, several behavioural and biological factors have been associated with increased BV risk, including vaginal douching, antibiotic exposure, intrauterine contraceptive device use, cigarette smoking, and unprotected sexual activity (17, 19). Additionally, emerging evidence suggests that BV prevalence and CST distribution may vary across different ethnic and geographical populations, highlighting the importance of considering population-specific microbial patterns in BV research.

2.1.2 Interspecies synergy, biofilm formation, and their contribution to BV persistence

The persistence and recurrence of BV are largely driven by complex interspecies interactions among BV-associated bacteria, which collectively facilitate biofilm formation and establish a polymicrobial community that is resilient to antimicrobial intervention (20). *Gardnerella* is a genus currently consisting of thirteen identified species following four decades of classification as a single species genus. Despite, this reclassification, the exact role of each *Gardnerella* spp in BV pathogenesis are poorly understood. Evidence suggests that *Gardnerella* spp. become pathogenic when multiple *Gardnerella* species co-occur, particularly with *G. vaginalis* (21, 22). *G. vaginalis* is detected in over 90% of BV cases (23), and is regarded as a primary colonizer of the vaginal epithelial cells (VECs), initiating BV pathogenesis through virulence traits. These include such as sialidases which degrade the sialoglycans present in vaginal mucus and vaginolysin, a pore-forming cytotoxin that disrupts VEC integrity, facilitating the subsequent attachment of *F. vaginae*, *P. bivia* and other BVABs to the established *G. vaginalis* scaffold. These polymicrobial attachments give rise to synergistic polymicrobial interactions that culminate in biofilm formation, a hallmark feature of BV. Biofilms are complex, three-dimensional structures that contribute to the increased virulence and persistence of BV. Their multilayered architecture facilitates the formation of polymicrobial co-aggregates within the distinct biofilm strata, thereby limiting antibiotic penetration and thereby reducing treatment efficacy. Biofilm development is a complex process initiated when planktonic cells irreversible attachment to the host epithelial surface. These aggregates subsequently develop into clumps that mature and disperse, allowing colonization of other locations on the host surface (24, 25).

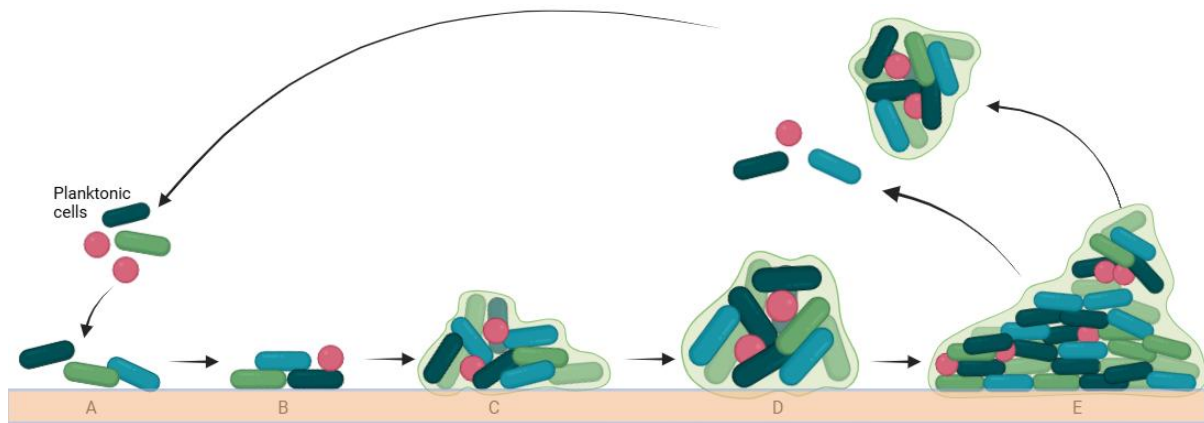


Figure 1: Formation of biofilms (25). The process begins with the initial attachment of cells to a substrate (A), followed irreversible attachment of cells (B), micro-colony formation (C) and biofilm maturation (D), and dispersion of cells or aggregates that recolonise other parts of the substrate (E). Created with Biorender.com

During BV, this polymicrobial biofilm functions as a protective scaffold sanctuary, encasing both planktonic cells and co-aggregates exacerbating persistence and recurrence. *In vitro* studies demonstrate a synergistic relationship within the multi-species biofilms involving *G. vaginalis*, *F. vaginae* and *P. bivia*. Although *F. vaginae* fails to form mono-species biofilms and does not substantially increase the total biomass (26), it proliferates in the presence of *G. vaginalis* and cell-free *G. vaginalis* supernatant, suggesting a commensal dependency, as *F. vaginae* is rarely detected in its absence (27). Interestingly, despite this dependency, *F. vaginae* independently mediates a pronounced pro-inflammatory response during BV pathogenesis (6). *P. bivia* similarly exploits *Gardnerella*-derived amino acids as preferential nutrient sources to enhance its own growth, while reciprocally producing ammonia that supports *G. vaginalis* proliferation. Moreover, *P. bivia* functions as a secondary colonizer with sialidases producing capability, thereby aiding in mucin degradation within *G. vaginalis*-driven biofilms (28). Collectively, these findings position *G. vaginalis* as a key driver of polymicrobial community dynamics in BV, that promote biofilm formation, ultimately, fostering interspecies dependencies that reduce treatment efficacy and promote recurrence.

2.1.3 Metronidazole as Standard Therapy for BV: Mechanism of Action

Metronidazole, popularly known as flagyl, is a synthetic drug consisting of a nitro group that was first discovered over six decades ago. MTZ is available and recommended in three

formulations (oral, intravenous and topical) (29). Unlike in aerobic bacteria where the prodrug MTZ remains inactive due to oxygen functioning as a final electron transport acceptor, in anaerobes, MTZ is activated in anaerobic organisms through the reduction of its nitro group by intracellular electron transport proteins, such as ferredoxin-linked nitroreductases. Upon penetrating the cell, the prodrug is converted into its active form, leading to toxic free radicals and hydroxylamine production that cause DNA strand breakage, and ultimately result in cell death (30).

During BV-associated dysbiosis, the vaginal microbiota is commonly dominated by *G. vaginalis*, *F. vaginalis* and *P. bivia*. However, other gram-negative anaerobes such as *Bacteroides fragilis* have also been isolated from vaginal swabs of women with BV (31). Although the exact mechanism of MTZ action against BVAB remains incompletely understood, its antimicrobial activity has been extensively reported in other anaerobes. Early *in vitro* studies demonstrated that MTZ rapidly eradicate *B. fragilis* more effectively than other antibiotics such as clindamycin. Additionally, MTZ activity against *B.* primarily involved rapid DNA synthesis inhibition without causing significant DNA structural damage or changes to the metabolic activity (32, 33). This phenomenon may partly reflect the metabolic adaptability and aerotolerance exhibited by *B. fragilis* under varying environmental conditions (34). Interestingly, although *E. coli* is classified as a facultative anaerobe, it has been shown to activate and accumulate MTZ under anaerobic conditions, thereby rendering it susceptible to the drug (35). The variation in the treatment effects among bacterial species may be attributed to different drug activation pathways, metabolic states, biofilm-forming capacity, and intrinsic antimicrobial resistance mechanisms employed by the microorganisms to survive the pressure exerted by antibiotics. Consequently, the polymicrobial nature of BV creates a complex microenvironment that promotes tolerant and resistant phenotypes, ultimately reducing clinical efficacy of MTZ treatment.

2.1.4 Metronidazole efficacy, recurrence, and the molecular mechanisms of treatment failure in BV

The eradication of BV using MTZ has been associated with high recurrence rates, with over 60% of women experiencing recurrence within 6-12 months (36). One proposed explanation for this poor long-term efficacy is the presence of structured polymicrobial biofilms, which limit antimicrobial penetration and contribute to bacterial persistence. Owing to the complex

three-dimensional multi-layer biofilm matrix, MTZ penetration into deeper biofilm layers is impaired, resulting in suboptimal drug exposure across the entire biofilm structure and reduced treatment effectiveness (37). BV recurrence rates are also perpetuated by failure to re-establish a stable *Lactobacillus* rich vaginal microbiota, particularly one enriched with protective species such as *L. crispatus*, *L. gasseri*, and *L. jensenii*. Instead, post-treatment vaginal communities are frequently dominated by *L.iners*, which are more likely to mediate the transition of the vaginal microbial community back into a BV state (38). Concerningly, current clinical management strategies do not adequately distinguish between BV relapse, persistence and re-infection, which complicates interpretation of treatment outcomes and recurrence patterns (37).

Research investigating the efficacy of MTZ on BV demonstrates that treatment non-responsiveness is multifactorial (39). Out of 14 taxa associated with increased HIV susceptibility, only *Gardnerella spp* showed resistance to MTZ (40). As previously described, *G. vaginalis* is the key driver of biofilm formation in BV pathogenesis, which is known to be resistant to MTZ. Similarly, a longitudinal study by Schwebs and colleagues evaluated resistant patterns using minimum inhibition concentration (MIC) of MTZ relative to CSLI and EUCAST guidelines across 3 visits: baseline, 6 weeks and 12 weeks. Only 30% of participants treated with standard dose of 2g of MTZ achieved initial cure at 6 weeks. The remaining 70% of the participants remained NS $\geq$ 3 and were retreated with 400mg/daily for 5 days (after 6 week visit). The *Gardnerella spp* MTZ MIC median increased progressively from baseline to week 6 visit, and significantly more from the 6 to 12 week-visit, indicating development of selection of increasingly resistant *Gardnerella spp* populations during treatment (17).

Furthermore, studies have demonstrated an increase in the relative abundance of *L. iners* following MTZ treatment, supporting observations by Andile Mtshali and colleagues that MTZ treatment frequently shifts the vaginal microbiota toward *L. iners* dominance rather than restoration of a protective *Lactobacillus*-rich state (17, 41). The abundance of *L. iners* before and after treatment has also been associated with BV due to its unstable ecological behaviour and high chances of transitioning towards CST IV communities (39, 42). Recurrent BV is suggested to be mediated by either the failure of MTZ to eradicate the persister cells or inability to re-establish a healthy vaginal microbiota composition dominated by *L. crispatus*, *L. gasseri* or *L. jensenii* (37). These findings imply that an interplay of community shifts of the vaginal microbiota, key BVAB resistance and potentially insufficient treatment regimens contribute to the high recurrence rates observed following MTZ therapy. While substantial research has

focused on determining the phenotypic susceptibility of BVAB to MTZ, studies investigating underlying genotypic and phenotypic mechanisms remain limited. Consequently, a significant knowledge gap exists regarding the molecular pathways that mediate BVAB survival during MTZ exposure, highlighting the need for mechanistic investigations using *in vitro* controlled models and clinically relevant MTZ concentrations.

Ambiguity between the different phenomena that influence resistant phenotypes such as tolerance and persistence exist as a result of the interchangeable use of the terms. Definitively, tolerance is a phenotype observed particularly within biofilms where bacteria survives transient exposure to fatal antibiotic concentrations, with adapted low growth rates (43). This phenotype is commonly associated with reduced growth rates and altered metabolic activity. Prolonged exposure to antimicrobials may lead to gene mutations therefore inducing polymicrobial adaptation and resistance. In contrast, persistence describes a transient dormant state adopted by a small subpopulation of bacterial cells within a tolerant community (25, 43). It is worth noting that, antimicrobial resistance is not always a sudden occurrence but rather an endpoint of polymicrobial interspecies-interactions, cooperation and adaptations as a measure to ensure survival in stressful environments. These interlinked phenotypes collectively enhance the resilience of polymicrobial communities during antimicrobial exposure (25).

2.1.5 Molecular mechanisms of MTZ resistance in anaerobic polymicrobial communities

Antibiotic resistance mechanisms in anaerobic bacteria include genes encoding nitroreductase enzymes, DNA repair, efflux pumps, all of which contribute to reduced MTZ efficacy by impairing drug activation, uptake, or intracellular retention. Biofilms are substantially more resistant to antibiotics as opposed to monospecies due to their polymicrobial organization and cooperative interactions. Polymicrobial biofilm survives unfavourable conditions through genotypic and phenotypic adaptations such as horizontal gene transfer of antibiotic resistant genes by conjugation, crossfeeding between interspecies, production of quorum sensing molecules, low growth rates and metabolic dormancy (43). These mechanisms collectively promote tolerance, persistence, and antimicrobial resistance within polymicrobial communities. The structure of the biofilm matrix serves as a mechanism that protects against environmental stressors such as antimicrobials. The matrix is composed of extracellular DNA polysaccharides, and proteins that maintain structural integrity, provide nutrients, and potentially reduce antimicrobial penetration through antibiotic binding or sequestration. This

structure, therefore reduces the rate of drug activation and ultimately results in sub-lethal concentrations reaching the cells (25).

Inadequate concentrations have been reported to select for resistant mutants and facilitate horizontal gene transfer of mobile genetic elements such as plasmids. Metabolic interactions such as cross feeding within polymicrobial species may further influence the retention of resistant plasmids. Studies have also demonstrated increase in the stability of resistance plasmid genes in polymicrobial biofilms than their planktonic counterparts (43, 44). An increase in cell biomass and target cell density within the mixed-species biofilm have improved conjugation efficiency in comparison to planktonic cells. Environmental stressors such as antimicrobials may mediate the upregulation of conjugation encoding genes, thereby facilitating the spread of antibiotic resistant genes. Moreover, conjugation transfer commonly occurs in gram negative polymicrobial biofilms and may facilitate the transfer of a number of genes associated with MTZ resistance (25, 44).

2.1.5.1 Nitroreductases, *nim* genes, and other resistance determinants

Nitroreductases play a central role in MTZ activation or inactivation (30). The downregulation of the *NfsA*-reductase like novel gene, *GvNRI*, has been demonstrated to induce direct MTZ resistance, in *G. vaginalis* through non-mutation-mediated mechanism (45). This response is typical of type-I nitroreductases, suggesting similar function. *Nim* genes encode resistance to 5-nitroimidazole drugs, including MTZ and are often located in chromosomes or plasmids which are horizontally transferred via conjugation or insertion elements (IS) capable of activating gene transcription (46, 47). Although the presence of *nim* genes does not directly translate to antibiotic resistance, in *Bacteriodes* spp, *nim* genes have been associated with MTZ resistance (48, 49). Of the *nim* genes described in *Bacteroides* spp (*nimA–H* and *nimJ*), *nimA–D* are associated with specific mobile genetic elements (MGEs), and pIP417 and pIP419-like plasmids, genetic determinants known to confer resistance against 5-nitroimidazole drugs (50). *NimK*, a novel gene encoded within an MGE (IS 1380 family transposase) was identified in three MTZ-resistant *P. bivia* strains harbouring the MGE. An associated small MDR efflux transporter was detected within the same genetic element (51), possibly suggesting that multiple mechanisms play a role in conferring MTZ resistance.

To date, only one study has reported the presence of a *nim* gene in *Gardnerella* spp. without investigating transcriptomic expression or functional activity, leaving the effects of *nim* gene presence on *Gardnerella* spp. resistance during MTZ treatment poorly characterised (52). Furthermore the limited surveillance of antimicrobial resistance genes in anaerobic bacteria remains a challenge (53), particularly in understanding resistance dynamics. An earlier study by Bostwick and colleagues employed next-generation sequencing in 26 BV positive and negative age-matched, non-pregnant women of reproductive age, to detect bacterial genes associated with resistance to a wide variety of antibiotics including macrolides, lincosamides, tetracyclines, aminoglycosides, 5-nitroimidazoles and triazoles. The AMR genes associated with 5-nitroimidazole resistance were the least abundant (0.3%) compared to macrolides (35.2%), lincosamides (35.6%), tetracyclines (21.8%), aminoglycosides (5.2%) and triazoles (18.7%). Furthermore a four fold AMR-gene increase in BV positive compared to BV negative samples was only identified in macrolides (58.2 versus 12.3%, respectively), lincosamides (58.9 versus 12.3%) and tetracyclines (35.6 versus 8.0%) (54). Similar findings were reported in a 2021 study, investigating presence of AMR genes in five BV positive women treated with MTZ, in which *nimJ* was detected at extremely low levels, while tetracycline resistance genes (*tetM* and *tetQ*) were the most abundant, and *lscA*, associated with clindamycin, lincomycin, dalbapristin, and tiamulin resistance, was the second most abundant (55). Despite the small sample size limitation, these studies suggest the need for whole-genome sequencing efforts that extend beyond *nim* gene presence as a marker of MTZ resistance to include additional MTZ resistance markers such as biofilm-formation associated genes, and stress response pathways.

2.1.5.2 DNA repair and efflux pump-mediated resistance

The RecA protein, encoded by the *recA* operon, plays a critical role in bacterial DNA repair and protects microbial DNA from MTZ-induced damage under low-redox conditions (46). Upon DNA damage mediated by ssDNA accumulation on the replication fork, the cells overexpress the SOS which is usually inhibited by *LexA* under normal DNA conditions. Studies suggest that the overexpression of *RecA* in *B. fragilis*, results in resistance to MTZ (51, 56), however this mechanism remains uncharacterized in key BVAB such as *G. vaginalis* and *F. vaginae*.

Efflux pump genes also play a major role in antibiotic resistance as a stress response mechanism. The upregulation of the *BmeRABC5* gene family in *B. fragilis* significantly

increased efflux pump activity, conferring resistance against MTZ (56), by reducing intracellular drug concentrations the effectiveness of antibiotics, particularly drugs that enter the cell via diffusion and require activation by nitroreductases. In the case of BV treatment, MTZ entry into the cells is combated in a number of ways, for example, as much as biofilm matrix hinders drug access into the biofilm, an intrinsic factor such as efflux pumps may exacerbate the ineffectiveness further by pumping out the antibiotics resulting in less optimal dosages acting on the target cells (52). While these mechanisms are established in anaerobic bacteria such as *B. fragilis* (51, 56), they are understudied in *G. vaginalis*, *F. vaginae* and *P. bivia*, which are key species that mediate BV pathogenesis as described by an updated conceptual model (57). Studies investigating the presence of genes encoding the different mechanisms of resistance in key bacteria involved in BV pathogenesis are needed to improve understanding of MTZ treatment failure and recurrent BV.

2.1.6 In Vivo Studies of BV Treatment Response: Insights and Limitations

A recent study evaluating recurrence in a South African cohort of BV positive women treated with MTZ, reported a low cure rate of 30% and further demonstrated that both *Gardnerella* sensitivity and resistant strains isolated from the same individuals persisted during MTZ treatment. This phenotype was hypothesized to be linked to resistant strains protecting the sensitive strains within biofilms (17), thereby highlighting the need to elucidate genomic signatures and strain variation analysis to identify drivers of persistence and recurrence. These findings further demonstrated incomplete efficacy suggesting that the vaginal condition following treatment remains unfavourable for successful regrowth of *Lactobacillus* (58).

Despite the important insight into resistance patterns and treatment outcomes provided by these clinical studies, the lack of mechanistic investigation remains a challenge in understanding the resistance pathways associated with BV-associated bacteria that contribute to treatment outcomes. Therefore, elucidating this phenomenon may play an important role in investigating drivers of treatment failure in BV (8). Collectively, the evidence from the under characterization of microbial determinants contributing to treatment failure in BV necessitates a robust controlled *in vitro* model that accurately mimics the *in vivo* environment as no two *in vivo* systems are the same. This limitation poses a challenge to mechanistic analysis. An *in vitro* model that incorporates the genomic signatures driving resistance and persistent BV

phenotypes would aid in identifying therapeutic targets and improving treatment strategies based on the resistome profile (55).

2.2 Problem Statement

Given that a highly diverse vaginal microbiome dominated by anaerobic bacteria results in adverse reproductive outcome in women of reproductive age and current clinical treatment is ineffective, while treatment failure is often attributed to biofilm formation and antimicrobial resistance, the mechanistic basis of these resistance phenotypes remains poorly understood. This is compounded by the polymicrobial complexity of BV, where interactions between multiple anaerobic species obscure the contribution of individual organisms to treatment failure. Furthermore, the inherent biological variability across individuals, including differences in vaginal microbiome composition influenced by ethnicity and menstrual cycle phase, limits the ability to identify consistent mechanistic patterns in clinical settings. Conventional diagnostic approaches such as Nugent scoring and Amsel criteria lack the resolution to detect subtle community shifts, and clinical studies rarely provide the longitudinal resolution necessary to capture real-time community dynamics during and after treatment. Collectively, these technical, ethical, biological, and diagnostic limitations underscore the need for a controlled *in vitro* BV model, to characterize resistance-associated genomic markers, and define the phenotypic determinants of metronidazole and clindamycin clearance, resistance and persistence.

2.3 Aims and Objectives

2.3.1 Aim

- To characterise microbial community shifts and identify molecular biomarkers associated with MTZ and clindamycin treatment success and failure in an *in vitro* model of BV.

2.3.2 Objectives

- To characterise the vaginal microbial community composition and dynamics before and after treatment using culturomics and molecular profiling approaches.
- To determine the genomic and molecular characteristics of bacterial taxa associated with treatment success and failure using metagenomic analyses.
- To compare microbial and molecular biomarkers associated with treatment outcomes identified *in vitro* with those observed *in vivo* among women treated for BV.

2.3.3 Hypothesis

- Distinct microbial community structures and molecular characteristics of key bacterial taxa drive treatment success and failure following MTZ and clindamycin therapy for BV.

3. METHODOLOGY

3.2.1 Study design, setting and sample collection

This study will utilise cryopreserved baseline vaginal samples collected from participants enrolled in the CAP098 cohort study to simulate BV-associated bacterial responses to sequential treatment *in vitro*. CAP 098 is a cohort study involving BV positive participants aged 18 -45 years that were recruited from the communities in the Durban vicinity and those attending the Prince Cyril Zulu Communicable Disease Centre and Lancer's Road clinics, which are public, open, and free-access HIV, TB and STI clinics. A regimen of 400 mg MTZ was administered twice daily for seven days and those who still remained BV positive were treated with clindamycin 2% intravaginal ointment. This study was conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants prior to any procedures. The CAP098 protocol was reviewed and approved by the University of KwaZulu-Natal Biomedical Medical Research Ethics Committee (BREC/00006875/2024). CAP098 participant vaginal samples were collected by the study clinicians at the different time points as stipulated by the study protocol. This study will conduct retrospective analysis on baseline vaginal swabs from some participants who tested BV cured (clearance) and BV positive (persistence and recurrence) after treatment regimens of the respective studies. Vaginal swabs for culturomics analysis were stored in thioglycolate storage media for culturomics analysis.

The *in vitro* model will recapitulate two distinct clinical treatment timelines. For CAP098-derived samples, the model will recapitulate either seven days of metronidazole alone, or seven days of metronidazole followed by seven days of clindamycin in women who were not cured following initial metronidazole treatment, with a subsequent two-day post-treatment recovery phase. Together, these timelines will enable modelling of a comprehensive spectrum of clinical outcome phenotypes, including clearance, partial response, persistence, and recurrence as observed in the CAP098 cohort.

This study will integrate three analytical components: the assembly of participant-derived polymicrobial consortia from clinical whole vaginal samples, simulation of antibiotic treatment sequences, and comprehensive phenotypic and genotypic characterisation of bacterial communities. To ensure representation of both dominant and contributing minor communities, whole vaginal swabs collected in thioglycolate broth will be incorporated into the *in vitro* model, preserving the anaerobic microbial diversity of the native vaginal environment.

3.3 Laboratory Procedures

3.3.1 Establishment of BV-like community

Two complementary community models will be established. Firstly, the baseline vaginal swab samples collected from women with confirmed BV by Nugent score or Amstel's criteria will be used to establish one of two complementary community BV models under strict anaerobic conditions (5% CO₂, 10% H₂ and 85% N₂).

To simulate BV model directly from the patient derived community, the vaginal swabs stored at -80 °C in thioglycolate broth enrichment to preserve anaerobic diversity will be thawed on ice and briefly vortexed. 50 µL of the thioglycolate sample will be added to 2.95 mL of S-Broth and cultured at 37 °C for 24 hours.

To simulate BV consortia, vaginal swabs samples stored in thioglycolate will be thawed, serially diluted with PBS, dispensed on agar plates (Blood, CNA and MRS) and incubated at 37°C for 48 to 72 hours. 10 colonies will be picked from each plate with distinct colonies, and the undiluted initial sample will be mixed in 50% glycerol in a 1:1 ratio (v/v) and stored in -80 °C (to preserve the initial patient-derived culture). The picked colonies will be re-streaked into fresh agar plates and incubated for 48 hours. A total of 3 re-streaking cycles will be done to isolate pure colonies. Colonies will be inoculated in S-broth and incubated at 37 °C for 24 hours. Glycerol stock will be prepared in several cryovials with 500 µL of 50% glycerol and 500 µL of cultures in s-broth, according to the number of colonies that were isolated per media type. The plates from which the colonies were picked, will be sealed using parafilm and sent for bacterial identification. Following identification, to achieve high quality biomass, colonies will be individually cultured in favourable media and anaerobic conditions in a pioreactor at the University of Cape Town, Institute of Infectious diseases and Molecular medicine.

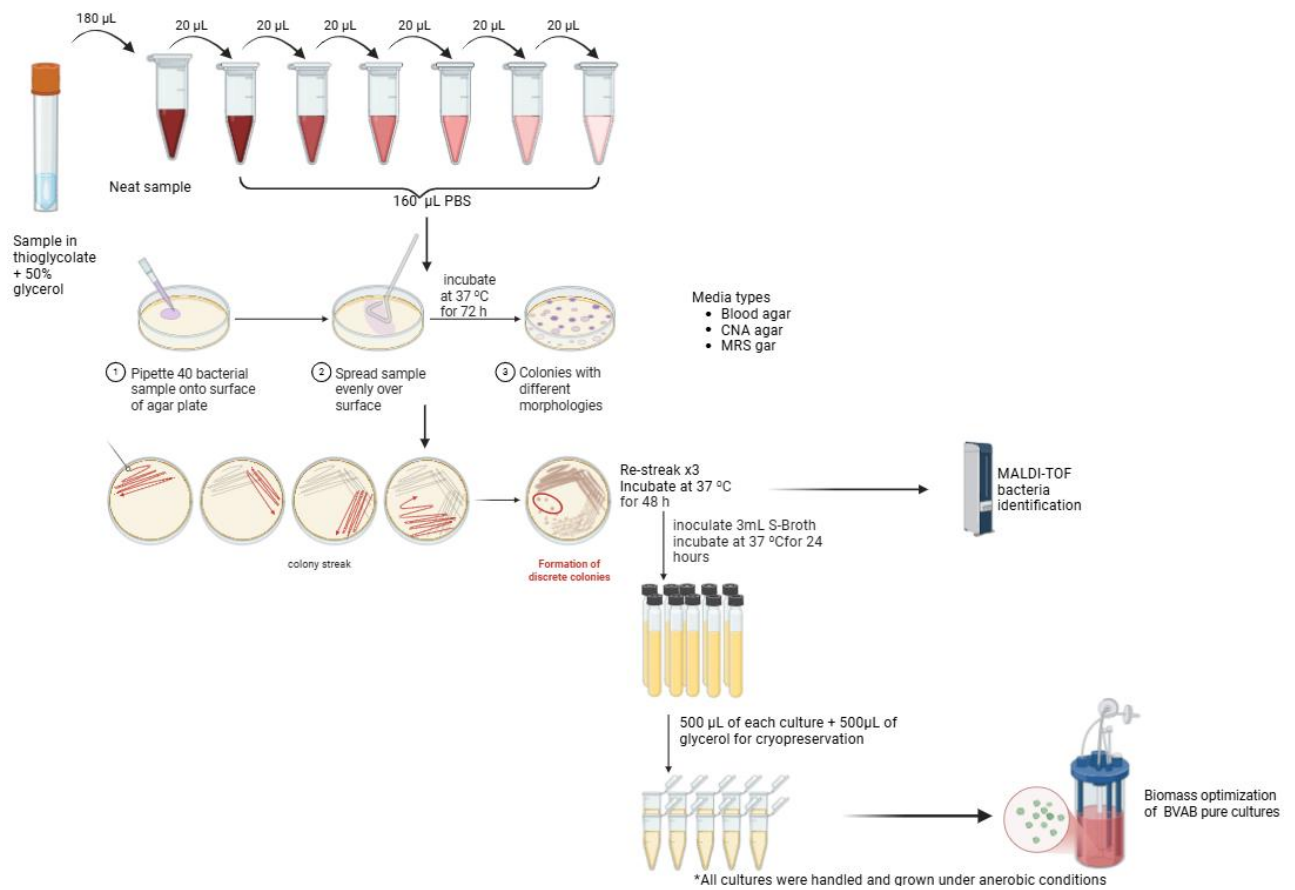


Figure 1: Workflow for obtaining anaerobic pure cultures for the establishment of BV consortia (competitive model)

To establish a competitive BV model, a defined equal amounts of key BV-associated species, including *G. vaginalis*, *A. vaginae*, *P. bivia*, and additional anaerobes biomass (e.g., *Mobiluncus spp.* where available) will be mixed to equal ratios in S-Broth incubated at 37 °C for 24 hours to establish a stable BV consortium. The mixed cultures will initially be cultured over 24 hours for stabilization before initiating treatment simulation.

3.3.2 Simulation of treatment and phenotypic assessment

Polymicrobial cultures will be exposed to clinically relevant concentrations of 30 µg/mL (based on tissue levels) of MTZ in S-Broth every 24 hours for 7 days, simulating systemic dosing, followed by clindamycin treatment every 24 hours for 7 days. A test-for-cure sample will be collected 48 hours post treatment course. Samples will be collected after every 24 hours for downstream analysis (16S rRNA PCR and qPCR) and glycerol storage. MTZ/clindamycin exposed cultures will be passaged every 24 hours.

3.3.2.1 Molecular and community shifts analysis

The qPCR and 16S rRNA sequencing will be used to validate the BV-models by assessing community composition at (baseline) and to track microbial community shifts during treatment exposure. Recurrence and persistence will be defined as regrowth of BV-associated taxa and or failure of *Lactobacillus* dominance post-MTZ/clindamycin treatment respectively.

3.3.2.2 DNA extraction

A semi-automated extraction will be performed using the DNeasy® PowerSoil® Pro Kit to extract genomic DNA from microbial sample and processed on the QIA cube Connect Instrument (Qiagen, USA?). The PowerBead Pro Tube will be spun briefly to ensure that beads settle at the bottom. Up to 250 mg of microbial sample and 800 µl of Solution CD1 will be added and briefly vortexed to mix. Samples in PowerBead Pro Tube (cat. no. 13000-V1-24) will be placed horizontally on a TissueLyser II (Qiagen, Germantown, MD, USA; #85300) and homogenised at 20 Hz for 20 minutes. Thereafter the PowerBead Pro Tube will be centrifuged at $15,000 \times g$ for 1 minute. Supernatant will be transferred to a clean 2 mL Microcentrifuge Tube. To further remove contaminating matter, 200 µl of Solution CD2 will be added, vortexed for 5 s and centrifuged at $15,000 \times g$ for 1 minute. Avoiding the pellet, 700 µl of supernatant will be transferred to a clean 2 ml Microcentrifuge Tube. 600 µl of Solution CD3 will be added and vortexed for 5 seconds for DNA binding to silica in the MB Spin Column filter membrane. 650 µl of the lysate will be loaded onto a MB Spin Column and centrifuged at $15,000 \times g$ for 1 minute. Flow-through will be discarded and centrifugation will be repeated to ensure that all the lysate has passed through the MB Spin Column.

Without splashing any flow through onto the MB Spin column, the column will be carefully placed into a clean 2 mL Collection Tube. The 500 µL of Solution EA will be added to the MB Spin Column and centrifuge at $15,000 \times g$ for 1 minute to remove protein and other contaminants. Flow-through will be discarded and the MB Spin Column will be placed back into the same 2 ml Collection Tube and 500 µl of Solution C5 will be added. The MB Spin Column will then be centrifuged at $15,000 \times g$ for 1 minute to further clean the DNA that is bound to the silica filter membrane in the MB Spin Column. The flow-through will be discarded and MB Spin Column will be placed into a new 2 ml Collection Tube and centrifuged at up to $16,000 \times g$ for 2 minutes. MB Spin Column will be carefully removed and placed into a new 1.5 ml Elution Tube to remove residual Solution C5. 50–100 µl of Solution C6 will be dispensed to the centre of the white filter membrane to wet the entire membrane for increased efficiency and release of the DNA from the MB Spin Column filter membrane. The MB Spin

Column in the Elution tube will be centrifuged at $15,000 \times g$ for 1 minute and the MB Spin Column will be discarded. DNA was quantified using Qubit 2.0 Fluorometer (Invitrogen, MA, USA). The DNA will be stored at -80°C and ready for downstream analysis (16S rRNA PCR, qPCR).

3.3.2.3 16S rRNA Amplicon Library preparation

Libraries will be prepared according to the Illumina 16S Metagenomic Library Prep guide (Document #15044223-B). DNA samples, positive controls (ZymoBIOMICS Microbial Community Standard, Zymo Research, Irvine, CA, USA; Cat#D6300) and negative controls (nuclease-free water, Qiagen, Cat#129114). The V3-V4 hypervariable region of the 16SrRNA gene will be targeted through a two-step amplification. For the first round of PCR, XT_338F and XT_806R gene specific primers (containing Illumina overhang adapter sequences and heterogeneity spacers 90-6bp) will be used to improve base diversity across reads. Seven primer variants per direction will be pooled in equimolar concentrations at $100\text{ }\mu\text{M}$ for stock and diluted to $10\text{ }\mu\text{M}$ for working solutions. This first step PCR reaction tubes will contain a $12.5\text{ }\mu\text{L}$ volumes of the Q5 High Fidelity DNA Polymerase (New England Biolabs, Ipswich, MA, USA; Cat #M0491S), $2.5\text{ }\mu\text{L}$ of $5 \times$ Q5 Reaction buffer (NEB, Cat#B9027S), $0.625\text{ }\mu\text{L}$ of each primer, $0.25\text{ }\mu\text{L}$ of dNTP mix (10Mm ; NEB, Cat#N0447S), $2.5\text{ }\mu\text{L}$ of template DNA and $5.875\text{ }\mu\text{L}$ of nuclease-free water. Amplification will be performed under the following thermocycling conditions; 98°C for 3 minutes, 30 cycles of 98°C for 30 seconds 58°C for 30 sec, 72°C for 7 minutes and a hold at 4°C .

For second-round PCR, unique dual index (UDI) primers with Nextera-compatible flow binding sequences and bp barcodes (Integrated DNA Technologies (IDT), Coralville, IA, USA, $100\text{ }\mu\text{M}$, 2 nmol scale) will be used. The $25\text{ }\mu\text{L}$ reaction will contain a $5\text{ }\mu\text{L}$ of $5 \times$ Q5 buffer, $2\text{ }\mu\text{L}$ index primer mix (forward and reverse), $0.5\text{ }\mu\text{L}$ dNTP mix ($10\text{ }\mu\text{M}$), $0.25\text{ }\mu\text{L}$ Q5 Polymerase, $2\text{ }\mu\text{L}$ undiluted first round PCR product, and $15.25\text{ }\mu\text{L}$ nuclease-free water. This PCR will be done under the following cycling conditions; 98°C for 1 minute, 10 cycles of 98°C for 30 sec, 60°C for 30 sec, and 68°C for 1 minute and thereafter followed by 68°C for 5 minutes and hold at 4°C . Agarose gel electrophoresis will be performed to confirm amplicon size ($\sim 627\text{ bp}$) using the GelRed nucleic acid stain (Biotium, Fremont, CA, USA; Cat#41003). After gel visualization, amplicon pooling will be done depending on the relative band intensity. Sample volumes will be taken as follows; $15\text{--}20\text{ }\mu\text{L}$ from weak bands, $7\text{ }\mu\text{L}$ from bands with normal intensity and $3\text{ }\mu\text{L}$ from samples with very intense bands. Amplicons will be pooled into a 2 mL tube, and the pool will be normalized to a volume of $100\text{ }\mu\text{L}$.

Purification of pooled amplicons will be done using the Agencourt AMPure XP PCR Purification Kit (Beckman Coulter, Brea, CA, USA; Cat#A63881). Prior to use, beads will be mixed. 60 µL of AMPure XP beads will be added for each 100 µL of pooled sample. The mixture will be homogenised by repetitive pipetting up and down 10 times. The mixture will be incubated at ambient temperature for 5 minutes to allow for DNA binding. Tubes will be placed on a magnetic rack for 10 minutes for solution clarification. Supernatant will be removed carefully to avoid bead loss. Beads will be washed with 200 µL of freshly prepared ethanol twice and incubated for 30s during each wash. Using a P10 multichannel pipette, residual ethanol will be removed, and beads will be air dried with open caps for 10 minutes. Elution will be performed using 50 µL of nuclease-free water mixed by pipetting. Magnetic separation will be done for 10 minutes, and eluate will be transferred to a new tube for quantification and sequencing. Fragment size and concentration of purified amplicon libraries will be verified using the Agilent 4150 Tape Station system (Agilent Technologies, Santa Clara, CA, USA: Cat#G2992AA) with the D5000 ScreenTape assay (Cat#5067-5592) and samples will be sent out for 16S rRNA sequencing.

3.3.2.4 Quantitative PCR

To quantify key culturable BV-associated bacteria and eubiotic bacteria during treatment exposure, quantitative PCR using specific Applied Biosystems™ TaqMan® assays for different taxa (*G. vaginalis* (Ba04646236_s1), *P. bivia* (Ba04646278_s1), *A. vaginae* (Pa04646150_s1), *Sneathia amnii*(APPRMPD) *Lactobacillus crispatus* (Ba04646245_s1), *Lactobacillus jensenii*(Ba04646258_s1), and *L. iners* (Ba04646257_s1)), will be performed on QuantStudio™ 5 Real-Time PCR System (Thermo Fisher Scientific, USA Cat. no. A32039). Each PCR reaction will contain a total volume of 10 µL that will contain, 5 µL TaqMan® Fast advanced Master Mix, 0.5 µL of each probe assay, 3.5 µL nuclease-free water and 1 µL of template DNA. Each of the targets will be quantified using amplicons generated from plasmids for each of the seven targets (TaqMan™ Vaginal Microbiota Extraction Control; cat no. A32039). A serial dilution from each plasmid will be used to determine the accurate quantification from the samples for the seven targets. Serial dilutions will range between 10 and 109 molecules per µL of sample (59).

3.3.2.5 Genotypic analysis

To identify antimicrobial resistance signatures of bacteria taxa identified as persistent by 16S rRNA analysis at key time points (baseline, post-MTZ, post-clindamycin and post treatment phase) will be re-cultured and samples will be sent for whole genome sequencing. Phenotypic outcomes (clearance vs persistence/recurrence) will be correlated with genomic and community profile. Multivariate analyses will be used to identify signatures predictive of treatment success or failure.

3.4 Statistical analysis

3.4.1 Differential abundance analysis of microbes

The differential abundance analysis of microbes between persistent/recurrent (BV positive) and susceptible (BV negative) samples will be performed by using the ANCOM-BC v2.0.3 R package (60). Indeterminant samples will be excluded. FDR-adjusted $P < 0.05$ and log2 fold-change (L2FC) ≥ 1 will be used to identify significantly enriched taxa. Significantly depleted taxa will identify with FDR-adjusted $P < 0.05$ and L2FC ≤ -1 . The remaining taxa will be identified as neutral (FDR-adjusted $P \geq 0.05$ or $-1 < \text{L2 FC} < 1$).

3.4.2 Community shifts and relative abundance

Samples susceptible and persistent post treatment will be compared with regards to various characteristics including the mean relative abundance and IAA of *Gardnerella* spp, *P. bivia*, *F. vaginae* and other common culturable BVAB (61). An unpaired t- test or Wilcoxon rank sum test, as appropriate will be used to compare continuous variables such as age. Sociodemographic, STI history, sexual behaviour history and contraception use at enrolment where applicable, will be compared between groups using χ^2 or Fisher's exact tests. Repeated measures analysis of variance (ANOVA) and linear mixed models will be used to examine longitudinal patterns in the sequencing data. The IAA of vaginal bacteria of interest, including *Gardnerella* spp, *P. bivia* and *F. vaginae*, will be obtained by multiplying each bacteria's relative abundance (measured by 16S rRNA gene sequencing) by their total bacterial load (measured by broad range 16S rRNA gene qPCR). Scatter plots and Pearson/Spearman correlation coefficients will be examine the relationship between the IAA and targeted qPCR absolute abundance of *Gardnerella* spp, *P. bivia* and *F. vaginae*. Bland-Altman method will be used to examine the agreement between the two measures. Continuous data in two time points will be compared by Wilcoxon matched-pairs signed-ranks test. All analyses will be conducted using R.

3.4.3 Genomic signatures of clearance and resistance

FastQC v. 0.11.82 will be used to assess the quality of the fastqc reads before and after quality trimming. Reads will be quality trimmed according to qualified_quality_phred of 20 and minimum read length of 50 using fastp v. 0.20.13. The reads will be human sequence depleted by aligning the trimmed reads to the human genome (hg38, University of California, Santa Cruz) using bowtie2 v. 2.3.4.14 with end-to-end alignment and a maximum fragment length for valid paired-end alignments (-X) of 2000 (62). Microbial profiling of the human DNA depleted reads will be performed with Kraken 2 v. 2.1.25 using the PlusPF database available from <https://benlangmead.github.io/aws-indexes/k2> which includes RefSeq genomes of archaea, bacteria, viruses, plasmids, human, UniVec Core, protozoa, and fungi. Species and genus level abundances will be estimated from the Kraken 2 reports using Bracken v. 2.6.2. The relative abundance of bacterial species and genera detected will be calculated relative to the total number of reads classified as bacterial. Species-level annotation will be performed with Metaphlan46, and gene and pathway functional annotation to MetaCyc will be performed with Humann37. Differential abundance analysis will be performed in R using the Aldex2 package⁸. The p-value considered will be from Welch's t-test.

SNV analysis will include all timepoints for all samples, where there will be adequate coverage to generate SNV profiles (62). MetaSNV under default settings, will be used to generate SNV profiles and reference databases will be generated using NCBI genomes. Bowtie2 will be used to align reads to the database and filtered to exclude alignments smaller than 45 base pairs and with less than 97% identity. Filtered reads will then be used to generate the SNV profile of each metagenomic sample and Manhattan distances will be calculated between profiles.

4. TIMELINE

Task	Feb 2026 - June 2026	June 2026- Aug 2026	Sep 2026 - Jan 2027	Feb 2027- Mar 2027	April 2027- May 2027	Jun 2027- October 2027
Proposal submission and defence						
BREC Application						
Laboratory analysis						
Data analysis						
Drafting results and discussion						
Final dissertation writing and submission						

5. REFERENCES

1. Amabebe E, Anumba DOC. The Vaginal Microenvironment: The Physiologic Role of Lactobacilli. *Frontiers in Medicine*. 2018;Volume 5 - 2018.
2. Dubé-Zinatelli E, Cappelletti L, Ismail N. Vaginal Microbiome: Environmental, Biological, and Racial Influences on Gynecological Health Across the Lifespan. *Am J Reprod Immunol*. 2024;92(6):e70026.
3. Mbulawa ZZA, Mabunda SA. Bacterial vaginosis associated with high rates of sexually transmitted infections among South African adolescent girls and young women. *Infection*. 2026;54(1):213–24.
4. Schuster HJ, Bos AM, Himschoot L, van Eekelen R, Matamoros SPF, de Boer MA, et al. Vaginal microbiota and spontaneous preterm birth in pregnant women at high risk of recurrence. *Heliyon*. 2024;10(10):e30685.
5. Spagnolo S, Tiffany K, Reynolds A, Zapata I. (038) RECURRENT BACTERIAL VAGINOSIS: PATIENT INSIGHTS ON MANAGEMENT AND CLINICIAN SUPPORT. *The Journal of Sexual Medicine*. 2025;22(Supplement_1).
6. Bradshaw CS, Plummer EL, Muzny CA, Mitchell CM, Fredricks DN, Herbst-Kralovetz MM, et al. Bacterial vaginosis. *Nat Rev Dis Primers*. 2025;11(1):43.
7. Mbaraka A, Meena RR, Menghani E, Verma N. Targeting biofilm-driven antibiotic resistance: emerging mechanisms and next-generation therapeutic interventions. *Frontiers in Microbiology*. 2026;Volume 17 - 2026.
8. Santos BPD, Gnoatto A, Heiden SM, Heimbecker V, de Carvalho NS, Marconi C. Systematic review of co-culture methods for studying vaginal biofilm formation in bacterial vaginosis. *J Microbiol Methods*. 2025;237:107243.
9. Condori-Catachura S, Ahannach S, Ticlla M, Kenfack J, Livo E, Anukam KC, et al. Diversity in women and their vaginal microbiota. *Trends in Microbiology*. 2025;33(11):1163–72.
10. Landolt EF, da Conceição Mendonça J, Behler AE, Lumsdaine SW, Jafar T, Burcham LR. Exploring the vaginal ecosystem: insights into host-microbe interactions and microbial community dynamics. *Infect Immun*. 2025;93(9):e0049924.
11. Bodaszewska-Lubaś M. Glycan Profile Alterations in Vaginitis and Their Role in Pathogen-Host Interactions. *Pol J Microbiol*. 2025;74(4):494–506.
12. Xu H, Zhang S, Zhang B, Jiang N, Xu Y, Chen X, et al. Vaginal colonization of Lactobacilli: Mechanism and function. *Microbial Pathogenesis*. 2025;198:107141.
13. Muzny CA, Łaniewski P, Schwebke JR, Herbst-Kralovetz MM. Host-vaginal microbiota interactions in the pathogenesis of bacterial vaginosis. *Curr Opin Infect Dis*. 2020;33(1):59–65.
14. Nugent RP, Krohn MA, Hillier SL. Reliability of diagnosing bacterial vaginosis is improved by a standardized method of gram stain interpretation. *Journal of Clinical Microbiology*. 1991;29(2):297–301.
15. Sanozky-Dawes R, Barrangou R. Lactobacillus, glycans and drivers of health in the vaginal microbiome. *Microbiome Res Rep*. 2022;1(3):18.
16. Ravel J, Gajer P, Abdo Z, Schneider GM, Koenig SSK, McCulle SL, et al. Vaginal microbiome of reproductive-age women. *Proceedings of the National Academy of Sciences*. 2011;108(supplement_1):4680–7.
17. Schwebs T, Kieninger AK, Podpera Tisakova L, Oberbauer V, Berdaguer R, Mtshali A, et al. Evaluation of Metronidazole Resistance of Vaginal Swab Isolates from South African Women Treated for Bacterial Vaginosis. *Antibiotics (Basel)*. 2024;13(12).

18. Jenkins DJ, Woolston BM, Hood-Pishchany MI, Pelayo P, Konopaski AN, Quinn Peters M, et al. Bacterial amylases enable glycogen degradation by the vaginal microbiome. *Nat Microbiol.* 2023;8(9):1641–52.
19. Ottinger S, Robertson CM, Branthoover H, Patras KA. The human vaginal microbiota: from clinical medicine to models to mechanisms. *Curr Opin Microbiol.* 2024;77:102422.
20. Vodstrcil LA, Muzny CA, Plummer EL, Sobel JD, Bradshaw CS. Bacterial vaginosis: drivers of recurrence and challenges and opportunities in partner treatment. *BMC Med.* 2021;19(1):194.
21. Klimowicz AK, Garcia EM, Jefferson KK, Dillard JP. Genetic transformation of *Gardnerella* species and characterization of vaginolysin and sialidase mutants. *bioRxiv.* 2025.
22. Munch MM, Strenk SM, Srinivasan S, Fiedler TL, Proll S, Fredricks DN. *Gardnerella* Species and Their Association With Bacterial Vaginosis. *J Infect Dis.* 2024;230(1):e171–e81.
23. Innamorati KA, Earl JP, Barrera SC, Ehrlich RL, Aiyeku J, Gordon A, et al. Metronidazole response profiles of *Gardnerella* species are congruent with phylogenetic and comparative genomic analyses. *Genome Med.* 2025;17(1):28.
24. Zhao A, Sun J, Liu Y. Understanding bacterial biofilms: From definition to treatment strategies. *Front Cell Infect Microbiol.* 2023;13:1137947.
25. Liu HY, Prentice EL, Webber MA. Mechanisms of antimicrobial resistance in biofilms. *npj Antimicrobials and Resistance.* 2024;2(1):27.
26. Castro J, Rosca AS, Muzny CA, Cerca N. *Atopobium vaginae* and *Prevotella bivia* Are Able to Incorporate and Influence Gene Expression in a Pre-Formed *Gardnerella vaginalis* Biofilm. *Pathogens.* 2021;10(2):247.
27. Castro J, Rosca AS, Cools P, Vaneechoutte M, Cerca N. *Gardnerella vaginalis* Enhances *Atopobium vaginae* Viability in an in vitro Model. *Frontiers in Cellular and Infection Microbiology.* 2020;Volume 10 - 2020.
28. George SD, Amerson-Brown MH, Sousa LGV, Rinehart AH, Tamhane A, Riegler AN, et al. Spatial organization of *Gardnerella* species, *Prevotella bivia*, and *Fannyhessea vaginalis* in the bacterial vaginosis biofilm. *Infect Immun.* 2026;94(2):e0063025.
29. Goolsby TA, Jakeman B, Gaynes RP. Clinical relevance of metronidazole and peripheral neuropathy: a systematic review of the literature. *International Journal of Antimicrobial Agents.* 2018;51(3):319–25.
30. Thomas C, Gwenin CD. The Role of Nitroreductases in Resistance to Nitroimidazoles. *Biology.* 2021;10(5):388.
31. Salman Z, Bunyan I, Umran B. Isolation and Identification of *Bacteroides fragilis* Isolated from Women with Bacterial Vaginosis in Hillah City. *Journal of Chemical and Pharmaceutical Sciences.* 2017;10:1619–22.
32. Sigeti JS, Guiney DG, Jr, Davis CE. Mechanism of Action of Metronidazole on *Bacteroides fragilis*. *The Journal of Infectious Diseases.* 1983;148(6):1083–9.
33. Connor B. Weir JKL. Metronidazole 2025 [cited 04 March 2026]. In: StatPearls [Internet]. Treasure Island (FL): StatsPearls Publishing, [cited 04 March 2026]. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK539728/>.
34. Spigaglia P. The Ambivalent Nature of *Bacteroides fragilis* and the Interaction with *Clostridioides difficile*: Benefits and Disadvantages for the Human Host. *Toxins (Basel).* 2025;17(10).

35. Jackson D, Salem A, Coombs GH. The in-vitro activity of metronidazole against strains of *Escherichia coli* with impaired DNA repair systems. *J Antimicrob Chemother.* 1984;13(3):227–36.
36. Qulu WP, Mzobe G, Mtshali A, Letsoalo MP, Osman F, San JE, et al. Metronidazole Treatment Failure and Persistent BV Lead to Increased Frequencies of Activated T- and Dendritic-Cell Subsets. *Microorganisms.* 2023;11(11).
37. Muzny CA, Sobel JD. Understanding and Preventing Recurring Bacterial Vaginosis: Important Considerations for Clinicians. *Int J Womens Health.* 2023;15:1317–25.
38. Abbe C, Mitchell CM. Bacterial vaginosis: a review of approaches to treatment and prevention. *Frontiers in Reproductive Health.* 2023;Volume 5 - 2023.
39. Nilsen T, Swedek I, Lagenaur LA, Parks TP. Novel Selective Inhibition of *Lactobacillus iners* by *Lactobacillus*-Derived Bacteriocins. *Appl Environ Microbiol.* 2020;86(20).
40. Valint DJ, Fiedler TL, Liu C, Srinivasan S, Fredricks DN. Effect of metronidazole on concentrations of vaginal bacteria associated with risk of HIV acquisition. *mBio.* 2024;15(12):e0111024.
41. Mtshali A, San JE, Osman F, Garrett N, Balle C, Giandhari J, et al. Temporal Changes in Vaginal Microbiota and Genital Tract Cytokines Among South African Women Treated for Bacterial Vaginosis. *Front Immunol.* 2021;12:730986.
42. Qin H, Jiao J, A D, Hua M, Han K, Du H, et al. Single-Molecule Approach to 16S rRNA for Vaginal Microbiome Signatures in Response to Metronidazole Treatment. *Microbiol Spectr.* 2023;11(3):e0170622.
43. Krewing M, Mönch E, Bolten A, Niesalla H. Resistance or tolerance? Highlighting the need for precise terminology in the field of disinfection. *Journal of Hospital Infection.* 2024;150:51–60.
44. Nithyanand P, Boya BR, Lee JH, Lee J. Polymicrobial Biofilms: Interkingdom Interactions, Resistance and Therapeutic Strategies. *Microb Biotechnol.* 2025;18(8):e70218.
45. Paunkov A, Mayr A-L, Oberbauer V, Schwebs T, Corsini L, Leitsch D. Downregulation of an NfsA-like nitroreductase causes metronidazole resistance in *Gardnerella vaginalis*. *International Journal of Antimicrobial Agents.* 2026;67(2):107681.
46. Alauzet C, Lozniewski A, Marchandin H. Metronidazole resistance and *nim* genes in anaerobes: A review. *Anaerobe.* 2019;55:40–53.
47. Muzny CA, Sobel JD. The Role of Antimicrobial Resistance in Refractory and Recurrent Bacterial Vaginosis and Current Recommendations for Treatment. *Antibiotics.* 2022;11(4):500.
48. Mahmood B, Juhász H, Leitsch D, Sóki J. The effects of identical *nim* gene-insertion sequence combinations on the expression of the *nim* genes and metronidazole resistance in *Bacteroides fragilis* strains. *Anaerobe.* 2023;81:102739.
49. Dingsdag SA, Hunter N. Metronidazole: an update on metabolism, structure–cytotoxicity and resistance mechanisms. *Journal of Antimicrobial Chemotherapy.* 2017;73(2):265–79.
50. Soki J, Gal M, Brazier J, Rotimi V, Urbán E, Nagy E, et al. Molecular investigation of genetic elements contributing to metronidazole resistance in *Bacteroides* strains. *The Journal of antimicrobial chemotherapy.* 2006;57:212–20.

51. Veloo ACM, Chlebowicz M, Winter HLJ, Bathoorn D, Rossen JWA. Three metronidazole-resistant *Prevotella bivia* strains harbour a mobile element, encoding a novel nim gene, nimK, and an efflux small MDR transporter. *J Antimicrob Chemother.* 2018;73(10):2687–90.
52. Younus N, Neela V, Gopinath R, Amin Nordin S. First report of nim gene encoding metronidazol in *Gardenerella vaginalis* 2013.
53. Ito Y, Hashimoto Y, Suzuki M, Kaneko N, Yoshida M, Nakayama H, et al. The emergence of metronidazole-resistant *Prevotella bivia* harboring nimK gene in Japan. *Microbiology Spectrum.* 2024;12(10):e00562–24.
54. Bostwick DG, Woody J, Hunt C, Budd W. Antimicrobial resistance genes and modelling of treatment failure in bacterial vaginosis: clinical study of 289 symptomatic women. *Journal of medical microbiology.* 2016;65(5):377–86.
55. Ruiz-Perez D, Coudray MS, Colbert B, Krupp K, Kumari H, Stebliankin V, et al. Effect of metronidazole on vaginal microbiota associated with asymptomatic bacterial vaginosis. *Access Microbiology.* 2021;3(5):000226.
56. Ghotaslou R, Bannazadeh Baghi H, Alizadeh N, Yekani M, Arbabi S, Memar MY. Mechanisms of *Bacteroides fragilis* resistance to metronidazole. *Infection, Genetics and Evolution.* 2018;64:156–63.
57. Muzny CA, Taylor CM, Swords WE, Tamhane A, Chattopadhyay D, Cerca N, et al. An Updated Conceptual Model on the Pathogenesis of Bacterial Vaginosis. *The Journal of Infectious Diseases.* 2019;220(9):1399–405.
58. Bloom SM, Symul L, Elsherbini J, Xu J, Hussain S, Shih J, et al. Microbiota effects and predictors of *Lactobacillus crispatus* colonization after treatment with a vaginal live biotherapeutic: results from a randomized, double-blinded, placebo-controlled trial. *medRxiv.* 2025.
59. Singh R, Ramsuran V, Mitchev N, Niehaus AJ, Han KSS, Osman F, et al. Assessing a diagnosis tool for bacterial vaginosis. *European Journal of Clinical Microbiology & Infectious Diseases.* 2020;39(8):1481–5.
60. Brochu HN, Zhang Q, Song K, Wang L, Deare EA, Williams JD, et al. Characterization of vaginal microbiomes in clinician-collected bacterial vaginosis diagnosed samples. *Microbiology Spectrum.* 2025;13(4).
61. Muzny CA, Elnaggar JH, Sousa LGV, Lima Â, Aaron KJ, Eastlund IC, et al. Microbial interactions among *Gardnerella*, *Prevotella* and *Fannyhessea* prior to incident bacterial vaginosis: protocol for a prospective, observational study. *BMJ Open.* 2024;14(2):e083516.
62. Wrønding T, Vomstein K, Lundgaard AT, DeLong K, Møllerup S, Mortensen B, et al. Vaginal microbiota transplantation for treatment of vaginal dysbiosis without the use of antibiotics: a double-blind, randomised controlled trial in women with vaginal dysbiosis. *The Lancet Microbe.* 2026;7(4):101294.